Notes: Morck, Yeung, and Yu (JFE, 2000)

The Information Content of Stock Markets: Why Do Emerging Markets Have Synchronous Stock Price Movements?

Contents

1 Big picture

This paper asks why stocks move together much more in emerging markets than in developed markets. The central empirical fact is simple: in countries such as China, Malaysia, Poland, Taiwan, and Turkey, a very high fraction of stocks move in the same direction in a typical week. In countries such as the United States, Canada, Ireland, Australia, and the United Kingdom, stock-level returns are much less synchronized.

The authors interpret this difference through the information content of stock prices. If many stocks move together, then a large share of return variation is market-wide. If stocks move more independently, then more return variation is firm-specific. Following Roll (1988), this distinction matters because firm-specific price movements are a natural proxy for firm-specific information being capitalized into prices.

Core message. Stock market synchronicity is not just a statistical property of returns. It is related to how well institutions protect property rights and how effectively stock markets process firm-specific information.

2 Incremental contribution of the paper

Main incremental contribution. The paper connects cross-country differences in stock return synchronicity to institutional quality, especially property rights protection, and interprets this as evidence about the information content of stock markets.

More specifically, the paper contributes in five ways.

1. **Documents a global synchronicity fact.** Poorer economies have more synchronous stock returns than richer economies. This is visible in raw weekly data, in country rankings, and in regression-based measures.

2. **Shows the pattern is not mechanical.** The result is not driven by the number of listed stocks, country size, market size, macroeconomic volatility, industry concentration, firm concentration, or measured earnings co-movement.

3. **Uses two complementary synchronicity measures.** The paper measures synchronicity both by the fraction of stocks moving in the same direction and by the share of stock return variation explained by market indexes.

4. **Links synchronicity to property rights.** Once the paper controls for a *good government* index, per capita GDP no longer explains synchronicity. This shifts the interpretation from income itself to institutions correlated with income.

5. **Distinguishes systematic versus firm-specific variation.** Emerging markets have especially high market-wide variation. Among developed markets, stronger shareholder protection is associated with more firm-specific variation.

Why this is new. Prior work emphasized whether markets are efficient in an absolute sense. This paper asks whether markets differ in the *type of information* embedded in prices. It links legal and political institutions to the amount of market-wide versus firm-specific information in stock prices.

3 Conceptual framework

3.1 Why synchronicity can be informative

Stock returns can move because of market-wide information or firm-specific information. If all stocks move together, then market-wide forces dominate. If individual stocks move differently, then firm-specific forces are important.

The paper evaluates three explanations for high stock return synchronicity in low-income economies:

1. **Correlated fundamentals.** Firms in low-income countries may be concentrated in similar industries, so their fundamentals may move together.
2. **Poor private property rights.** Political risk, expropriation risk, contract repudiation risk, and corruption may produce market-wide price shocks unrelated to firm fundamentals. Poor property rights may also deter informed arbitrage.
3. **Weak public investor protection.** If corporate insiders can shift resources across firms, firm-specific accounting information becomes less useful, reducing the incentive to trade on firm-specific information.

The paper finds little support for the first explanation as a complete account. It finds more support for the second and third explanations.

3.2 Mechanism

The proposed mechanism is as follows. Strong property rights allow informed arbitrageurs to earn and keep the returns to gathering firm-specific information. That makes firm-level information more likely to be incorporated into prices. Weak property rights make informed trading less attractive and create space for market-wide noise trader risk or politically driven price movements.

Interpretation. In this view, low synchronicity is a sign that many stocks contain different firm-specific information. High synchronicity is a warning that market prices may be reacting to broad political or noise shocks rather than detailed firm news.

4 Data and measurement

4.1 Data

The main cross-country analysis uses Datastream stock returns for 40 countries in 1995. Returns are dividend-adjusted and trimmed at $\pm 25\%$. The paper also uses country characteristics such as per capita GDP, number of listed stocks, geographical area, GDP growth volatility, industry Herfindahl measures, firm Herfindahl measures, and institutional variables from La Porta et al. datasets.

The paper also studies the United States over the twentieth century using Center for Research in Security Prices data. This U.S. time series shows that stock price synchronicity fell over time as the market developed.

4.2 Synchronicity measure 1: fraction of stocks moving together

For country j and period t, define

$$f_{jt} = \frac{\max(n_{jt}^{up}, n_{jt}^{down})}{n_{jt}^{up} + n_{jt}^{down}},$$

where n_{jt}^{up} is the number of stocks whose prices rise and n_{jt}^{down} is the number whose prices fall. The country-level measure is

$$f_j = \frac{1}{T} \sum_t f_{jt}.$$

A value close to 0.5 means roughly equal numbers of stocks rise and fall. A value close to 1 means almost all stocks move together.

4.3 Synchronicity measure 2: market-model R^2

The second measure is based on stock-level regressions of the form

$$r_{it} = \alpha_i + \beta_{1,i} r_{m,jt} + \beta_{2,i}(r_{US,t} + e_{jt}) + \varepsilon_{it},$$

where $r_{m,jt}$ is the domestic market return and $r_{US,t} + e_{jt}$ is the U.S. market return translated into local currency.

The paper defines a country-level R_j^2 as the average fraction of stock return variation explained by the local and U.S. market returns. A high R_j^2 means stocks are moving strongly with market-wide factors.

4.4 Logistic transformations

Because f_j and R_j^2 are bounded variables, the paper transforms them before using them as dependent variables:

$$\Psi_j = \log\left(\frac{f_j - 0.5}{1 - f_j}\right),$$

$$\Upsilon_j = \log\left(\frac{R_j^2}{1 - R_j^2}\right).$$

These transformed variables are then regressed on GDP, market size, structural controls, and institutional variables.

4.5 Institutional measures

The key institutional variable is the **good government index**. It is the sum of three indexes, each ranging from zero to ten:

- government corruption;
- risk of expropriation of private property;

- risk of government repudiation of contracts.

Higher values mean stronger respect for private property rights. The paper also uses measures of accounting standards and anti-director rights to study investor protection, especially within developed economies.

5 Empirical design

The main cross-country regression has the form

$$\Psi_j \text{ or } \Upsilon_j = c_0 + c_1 \log(y_j) + c_2 \log(n_j) + c' x_j + u_j,$$

where y_j is per capita GDP, n_j is the number of listed stocks, and x_j is a vector of structural variables.

To test the institutional explanation, the paper adds the good government index:

$$\Psi_j \text{ or } \Upsilon_j = c_0 + c_1 \log(y_j) + c_2 \log(n_j) + c' x_j + c_3 g_j + u_j.$$

The key test is whether the good government index is negative and significant, and whether adding it makes per capita GDP insignificant. That is what the paper finds.

5.1 Variance decomposition

The paper also decomposes return variation into systematic and firm-specific components. Let

$$\sigma_j^2 = \sigma_{m,j}^2 + \sigma_{\varepsilon,j}^2,$$

where $\sigma_{m,j}^2$ is market-wide variation and $\sigma_{\varepsilon,j}^2$ is firm-specific variation.

This decomposition asks whether high synchronicity comes from more systematic risk, less firm-specific information, or both.

6 Main results

1. **Emerging markets have high synchronicity.** The average fraction of stocks moving in the same direction is much higher in emerging markets. The market-model R^2 is also much higher.
2. **The United States became less synchronous over time.** Early twentieth-century U.S. stock returns looked more like today's emerging markets. U.S. synchronicity declined over the twentieth century.
3. **Structural controls do not explain the income-synchronicity relation.** Per capita GDP remains significant after controls for number of stocks, country size, GDP volatility, industry structure, firm concentration, and earnings co-movement.
4. **Good government explains the relation better than income.** Once the good government index is included, per capita GDP becomes insignificant and the good government index is negatively related to synchronicity.

5. **High synchronicity in weak-institution countries comes mainly from high systematic variation.** Market-wide variation is especially high in countries with weak property rights.

6. **Among developed economies, shareholder protection matters for firm-specific variation.** Stronger anti-director rights are associated with greater firm-specific return variation.

Economic message. Stock prices in developed markets appear to contain more firm-specific information. Stock prices in emerging markets appear more dominated by market-wide shocks that may be political, institutional, or noise-driven.

7 Identification and interpretation

The paper is not a natural experiment. It is a cross-country empirical design, so its interpretation relies on comparing alternative explanations and robustness checks.

Strengths of the design.

- The raw fact is large and visible across multiple synchronicity measures.
- The paper checks whether the result is merely due to market size, economy size, macro volatility, industry concentration, firm concentration, or earnings co-movement.
- The U.S. time-series evidence supports the idea that synchronicity declines as markets and institutions develop.
- The institutional measures have the right sign and make per capita GDP lose explanatory power.

Main limitations.

- Institutional quality is correlated with many aspects of development, so causal interpretation is difficult.
- Cross-country samples are small, especially after adding accounting and investor-rights variables.
- Fundamentals co-movement is measured imperfectly; omitted fundamental risks could still matter.
- The paper's mechanism - stronger property rights promote informed arbitrage - is plausible but not directly observed.

Best interpretation. The paper should be read as strong evidence that stock return synchronicity is systematically related to institutional development, and as suggestive evidence that weak property rights reduce the firm-specific information content of stock prices.

8 Tables and figures

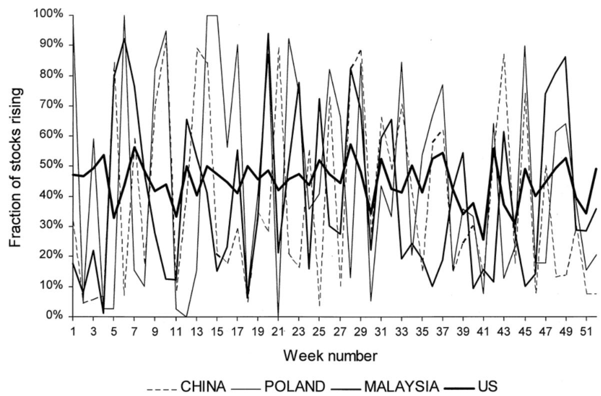
8.1 Table 1 and Figure 1: the motivating synchronicity fact

Table 1
Stock price comovement in selected emerging and developed stock markets

The fraction of stocks whose prices go up, go down, and remain the same during each of the first 26 weeks of 1995. Price changes are from Datastream, and are adjusted for dividends.

	China			Malaysia			Poland			Denmark			Ireland			United States		
Week	%Up	%Down	%Same	%Up	%Down	%Same	%Up	%Down	%Same	%Up	%Down	%Same	%Up	%Down	%Same	%Up	%Down	%Same
	32	61	7	18	73	9	97	3	0	50	29	21	39	46	16	47	29	24
2	4	89	6	8	86	6	5	95	0	45	25	30	33	32	35	47	38	15
3	6	88	7	22	69	9	59	31	10	36	33	31	32	40	28	49	37	13
4	7	88	5	1	95	3	3	92	5	27	36	37	33	32	35	54	32	14
5	84	8	7	80	11	9	3	97	0	48	33	18	44	26	30	33	53	15
6	7	50	42	92	2	6	100	0	0	41	30	29	42	39	19	44	43	14
7	59	31	10	77	14	10	15	77	8	41	30	28	42	40	18	57	30	13
8	18	73	9	47	39	13	10	90	0	29	35	36	28	35	37	48	38	14
9	71	22	7	28	60	12	82	13	5	40	33	27	37	42	21	42	43	15
10	93	4	4	13	77	11	95	5	0	23	36	41	25	30	46	44	42	14
11	9	88	3	12	78	9	3	95	3	31	38	31	26	39	35	33	52	15
12	41	51	7	66	23	11	0	92	8	30	37	33	28	39	33	50	37	13
13	89	7	4	53	34	13	15	67	18	21	36	42	35	39	26	41	44	15
14	84	9	6	41	50	8	100	0	0	28	37	35	32	44	25	50	35	15
15	21	73	5	15	73	12	100	0	0	27	43	30	33	39	28	47	37	15
16	18	75	7	23	66	11	56	38	5	30	52	18	28	46	26	45	40	15
17	29	63	8	56	25	19	90	10	0	34	40	26	42	37	21	41	44	15
18	5	92	3	6	87	6	8	92	0	48	33	18	47	37	16	50	35	15
19	35	56	9	33	57	10	41	49	10	39	36	26	35	44	21	46	40	14
20	29	60	11	94	3	3	87	10	3	41	36	22	40	35	25	49	37	14
21	89	8	3	21	72	7	0	100	0	39	35	26	46	37	18	42	44	14
22	21	76	4	51	42	7	92	5	3	38	33	29	40	44	16	46	39	15
23	16	79	5	78	17	5	74	23	3	34	40	26	49	44	07	47	39	14
24	55	37	8	16	77	7	36	51	13	24	40	36	40	33	26	44	41	15
25	4	84	12	72	18	9	41	49	10	22	41	37	49	33	18	52	34	14
26	73	20	7	30	60	9	82	5	13	26	40	34	39	49	12	47	39	14
Sample	308 stocks			349 stocks			38 stocks			233 stocks			57 stocks			6,889 stocks		

Stock price comovement in selected emerging and developed markets.



ig. 1. Observed stock price synchronicity in the stock markets of selected countries The fraction of tocks whose prices rise each week of 1995 in the stock markets of China, Malaysia, Poland and the Jnited States based on returns including dividend income from Datastream.

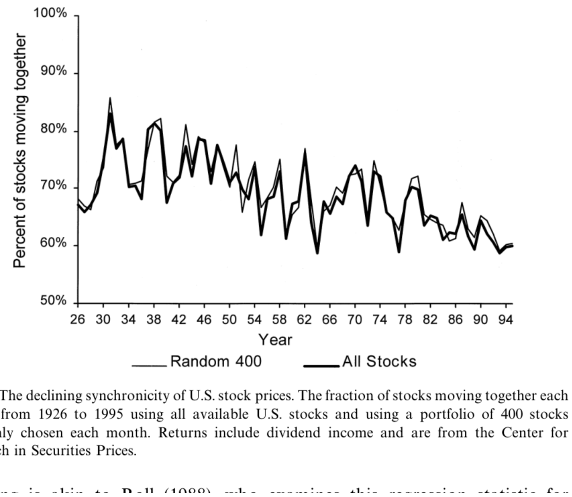
Observed stock price synchronicity in selected countries.

Read:

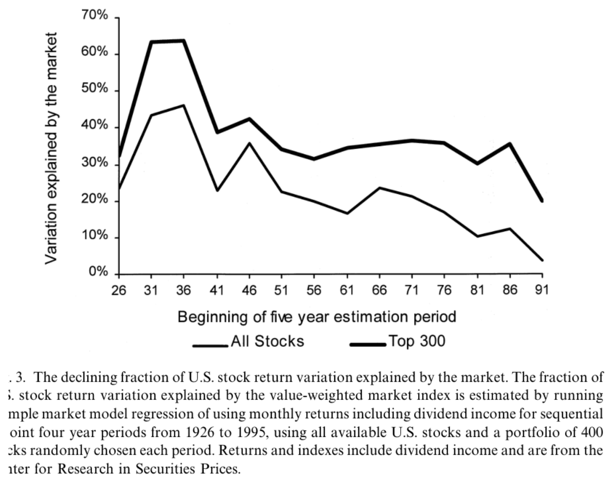
- Table 1 shows weekly up/down/same movements for China, Malaysia, Poland, Denmark, Ireland, and the United States during the first 26 weeks of 1995.
- Figure 1 makes the contrast visual: China, Malaysia, and Poland have many weeks in which most stocks move in the same direction; the United States is much less synchronized.

Economic message: The paper begins with a fact that is too large to dismiss as sampling noise: stock prices in many emerging markets move together dramatically more than U.S. stock prices.

8.2 Figures 2 and 3: U.S. synchronicity over the twentieth century



Declining synchronicity of U.S. stock prices.



Declining U.S. return variation explained by the market.

Read:

- Figure 2 shows that the fraction of U.S. stocks moving together falls from the early twentieth century to the 1990s.
- Figure 3 shows the same idea using market-model R^2: the market explained a much larger fraction of U.S. stock return variation in the 1930s than in later decades.

Economic message: Today's developed markets did not always look like developed markets. The U.S. historical pattern supports the idea that market development is associated with less synchronous, more firm-specific stock price movements.

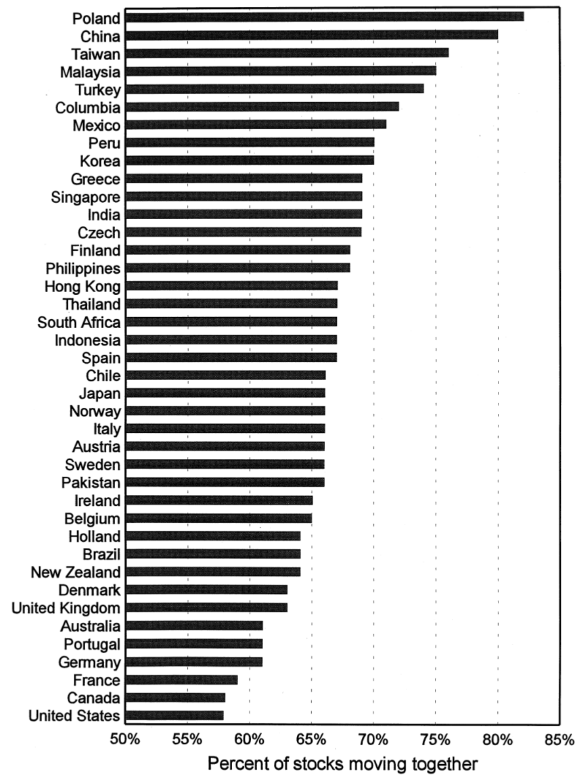
8.3 Table 2 and Figure 4A: country rankings by synchronicity

Table 2
Per capital gross domestic product and stock return synchronicity

Countries are ranked by per capita GDP in Panel A. In panel B, countries are ranked by stock return synchronicity, measured by the fraction of stocks moving together in the average week of 1995. In panel C, countries are ranked by stock market synchronicity, measured as the average R^2 of firm-level regressions of bi-weekly stock returns on local and U.S. market indexes in each country in 1995. Returns include dividends and are trimmed at $\pm 25\%$.

Panel A			Panel B		Panel C	
Country	Number of listed stocks	1995 per capita US$ GDP	Country	% stocks moving in step (f_j)	Country	R_j^2
Japan	2276	33,190	United States	57.9	United States	0.021
Denmark	264	27,174	Canada	58.3	Ireland	0.058
Norway	138	25,336	France	59.2	Canada	0.062
Germany	1232	24,343	Germany	61.1	U.K.	0.062
United States	7241	24,343	Portugal	61.2	Australia	0.064
Austria	139	23,861	Australia	61.4	New Zealand	0.064
Sweden	264	23,861	U.K.	63.1	Portugal	0.068
France	982	23,156	Denmark	63.1	France	0.075
Belgium	283	21,590	New Zealand	64.6	Denmark	0.075
Holland	100	20,952	Brazil	64.7	Austria	0.093
Singapore	381	20,131	Holland	64.7	Holland	0.103
Hong Kong	502	19,930	Belgium	65.0	Germany	0.114
Canada	815	19,149	Ireland	65.7	Norway	0.119
Finland	104	18,770	Pakistan	66.1	Indonesia	0.140
Italy	312	18,770	Sweden	66.1	Sweden	0.142
Australia	654	17,327	Austria	66.2	Finland	0.142
U.K.	1628	17,154	Italy	66.6	Belgium	0.146
Ireland	70	14,186	Norway	66.6	Hong Kong	0.150
New Zealand	137	12,965	Japan	66.6	Brazil	0.161
Spain	144	12,965	Chile	66.9	Philippines	0.164
Taiwan	353	10,698	Spain	67.0	Korea	0.172
Portugal	90	9,045	Indonesia	67.1	Pakistan	0.175
Korea	461	7,555	South Africa	67.2	Italy	0.183
Greece	248	7,332	Thailand	67.4	Czech	0.185
Mexico	187	3,944	Hong Kong	67.8	India	0.189
Chile	190	3,361	Philippines	68.8	Singapore	0.191
Malaysia	362	3,328	Finland	68.9	Greece	0.192
Brazil	398	3,134	Czech	69.1	Spain	0.192
Czech	87	3,072	India	69.5	South Africa	0.197
South Africa	93	2,864	Singapore	69.7	Columbia	0.209
Turkey	188	2,618	Greece	69.7	Chile	0.209
Poland	45	2,322	Korea	70.3	Japan	0.234
Thailand	368	2,186	Peru	70.5	Thailand	0.271
Peru	81	1,920	Mexico	71.2	Peru	0.288
Columbia	48	1,510	Columbia	72.3	Mexico	0.290
Philippines	171	880	Turkey	74.4	Turkey	0.393
Indonesia	218	735	Malaysia	75.4	Taiwan	0.412
China	323	455	Taiwan	76.3	Malaysia	0.429
Pakistan	120	424	China	80.0	China	0.453
India	467	302	Poland	82.9	Poland	0.569

Per capita GDP and stock return synchronicity.



.. Stock price synchronicity in various countries. Panel A. Stock price synchronicity measured e average fraction of stock prices moving in the same direction during an average week in 1995. prices that do not move during a week are excluded from the average for that week. Price ments are adjusted for dividend payments, and are based on Datastream total returns. Panel B. price synchronicity measured by the average percent of total biweekly firm-level return tion in 1995 explained by local and U.S. value-weighted market indexes. Stock returns and es include dividend payments, and are obtained from Datastream.

Stock price synchronicity by fraction moving together.

Read:

- Table 2 ranks countries by income, by the fraction of stocks moving in step, and by market-model R^2.
- Figure 4A ranks countries by the average fraction of stocks moving together. Poland, China, Taiwan, Malaysia, and Turkey are at the top; the United States, Canada, France, Germany, and Portugal are near the bottom.

Economic message: The paper's main cross-country pattern is visible before any regression: richer countries tend to have more asynchronous stock prices.

8.4 Figure 4B and Figure 5: market R^2 and income scatterplots

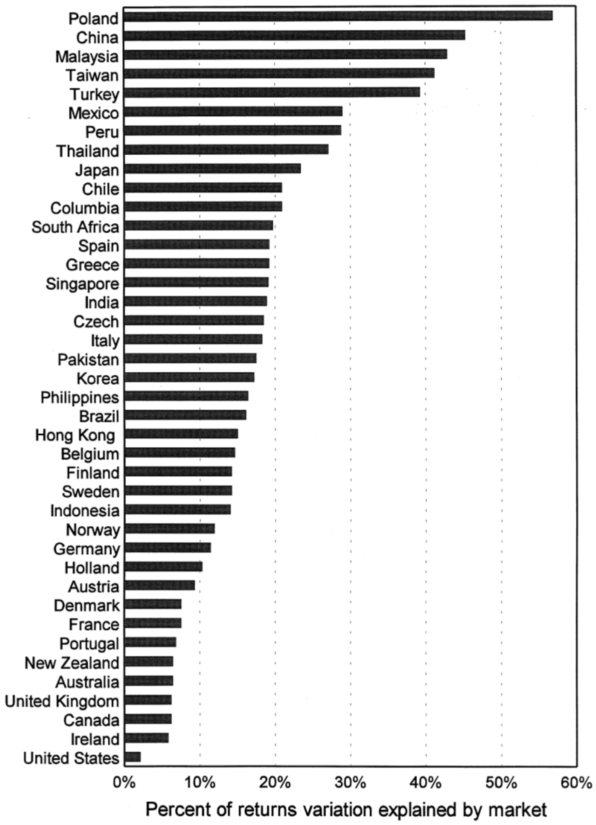


Fig. 4 (*continued*).

. 5. Stock price synchronicity and gross domestic product. Panel A. The logarithm of per capita ss domestic product plotted against stock return synchronicity measured by the average percent tock returns moving in the same direction each week. Each observation is for one country. Data

Stock price synchronicity by market-model R^2.   Stock price synchronicity and per capita GDP.

Read:

- Figure 4B ranks countries by the average percent of return variation explained by local and U.S. market indexes.
- Figure 5 plots the log of per capita GDP against both synchronicity measures. The scatterplots show two broad clusters: high-income/low-synchronicity countries and low-income/high-synchronicity countries.

Economic message: The R^2 measure tells the same story as the simpler up/down measure. Emerging markets have more market-wide price movement.

8.5 Table 3, parts 1 and 2: variables and simple correlations

Table 3
Description of main variables

Univariate statistics and simple correlation coefficients between main variables. Sample is 37 countries, except for the accounting standards index, which is available for 34 countries and the earnings co-movement index, which is available for only 25 counties. Numbers in parenthesis are probability levels at which the null hypothesis of zero correlation can be rejected in two-tailed tests.

Panel A. Univariate statistics and simple correlation coefficients between stock price synchronicity indices, stock return variance decomposition variables $\log(\sigma_\varepsilon^2)$ and $\log(\sigma_m^2)$, and structural and institutional variables.

					Simple correlation with			
Variables	Mean	Standard deviation	Minimum	Maximum	Ψ_i	Υ_j	$\log(\sigma_\varepsilon^2)$	$\log(\sigma_m^2)$
Stock co-movement indices								
Average fraction of stocks moving the same direction (f_j)	0.659	0.052	0.570	0.772	0.993 (0.00)	0.900 (0.00)	0.162 (0.34)	0.855 (0.01)
R square of market model based on weekly data for country j	0.169	0.099	0.020	0.429	0.888 (0.00)	0.949 (0.00)	0.146 (0.39)	0.891 (0.00)
Logistic transformation of f_j for country j (Ψ_j)	– 0.808	0.501	– 1.84	0.180	—			
Logistic transformation of R^2 for country j (Υ_j)	– 1.76	0.758	– 3.84	– 0.284	0.909 (0.00)	—		
Logarithm of firm-specific variation ($\log(\sigma_\varepsilon^2)$)	– 2.30	0.360	– 3.05	– 1.52	0.115 (0.50)	0.073 (0.67)	—	
Logarithm of market-wide variation ($\log(\sigma_m^2)$)	– 3.97	0.930	– 5.60	– 1.86	0.843 (0.00)	0.904 (0.00)	0.449 (0.00)	—

Main variables and simple correlations, part 1.

Logarithm of per capita GDP	8.94	1.30	5.71	10.4	– 0.512 (0.00)	– 0.457 (0.00)	– 0.406 (0.01)	– 0.573 (0.00)
Logarithm of number listed stocks	5.61	1.06	3.81	8.89	– 0.381 (0.02)	– 0.307 (0.06)	0.200 (0.23)	– 0.183 (0.28)
Structural variables								
Logarithm of geographical size	12.7	2.11	6.46	16.1	– 0.160 (0.34)	– 0.0105 (0.54)	0.372 (0.02)	0.084 (0.62)
Variance in GDP growth	0.0001	0.0002	0.0007	0.001	0.0703 (0.68)	0.0999 (0.56)	– 0.190 (0.26)	0.010 (0.97)
Industry Herfindahl index	0.113	0.0559	0.030	0.281	0.0116 (0.94)	– 0.035 (0.84)	– 0.0175 (0.28)	– 0.020 (0.00)
Firm Herfindahl index	0.0482	0.0505	0.0001	0.219	– 0.001 (0.99)	– 0.126 (0.46)	– 0.142 (0.38)	– 0.148 (0.36)
Earnings co-movement index	0.383	0.164	0.055	0.777	0.0555 (0.80)	0.201 (0.35)	0.100 (0.63)	0.250 (0.23)
Institutional variables								
Good government index	23.9	4.98	12.9	29.6	– 0.552 (0.00)	– 0.527 (0.00)	0.477 (0.00)	0.664 (0.00)
Accounting standards index	63.7	10.9	0.36	83	– 0.237 (0.18)	– 0.230 (0.19)	– 0.034 (0.85)	– 0.218 (0.22)
Anti-director rights index	1.78	1.93	0	5	– 0.586 (0.00)	– 0.595 (0.00)	0.271 (0.10)	– 0.077 (0.65)

Main variables and simple correlations, part 2.

Read:

- Table 3 reports summary statistics and simple correlations for synchronicity measures, income, market size, structural variables, and institutional variables.
- The simple correlations already suggest that synchronicity is negatively related to income and to the good government index.

Economic message: The institutional story is not only visible in multivariate regressions. It is already apparent in the raw cross-country correlation patterns.

8.6 Table 3 part 3 and Table 4: structural explanations

Table 3 (*continued*)

Panel B. Simple correlation coefficients of structural and institutional variables with each other

	a	b	c	d	e	f	g	h	i
a. Logarithm of per capita GDP	—								
b. Logarithm of number of stocks listed	0.364 (0.03)	—							
Structural variables									
c. Logarithm of geographical size	– 0.371 (0.02)	0.111 (0.51)	—						
d. Variance in GDP growth	– 0.020 (0.91)	– 0.196 (0.24)	0.010 (0.97)	—					
e. Industry Herfindahl index	0.025 (0.88)	– 0.674 (0.00)	– 0.214 (0.20)	0.115 (0.97)	—				
f. Firm Herfindahl index	– 0.020 (0.91)	– 0.573 (0.00)	– 0.040 (0.82)	0.091 (0.59)	0.710 (0.00)	—			
g. Earnings Co-movement index	– 0.03 (0.88)	0.105 (0.63)	0.109 (0.61)	– 0.100 (0.64)	– 0.168 (0.43)	– 0.325 (0.12)	—		
Institutional variables									
h. Good government Index	0.919 (0.00)	0.335 (0.04)	– 0.298 (0.07)	– 0.010 (0.96)	– 0.040 (0.82)	0.011 (0.95)	– 0.126 (0.56)	—	
i. Anti-director rights	0.706 (0.00)	0.403 (0.01)	– 0.259 (0.12)	– 0.292 (0.08)	– 0.060 (0.73)	– 0.070 (0.65)	– 0.108 (0.61)	0.729 (0.00)	—
j. Accounting standards index	0.442 (0.01)	0.427 (0.01)	– 0.090 (0.60)	– 0.265 (0.13)	– 0.552 (0.00)	– 0.267 (0.13)	0.035 (0.87)	0.554 (0.00)	0.531 (0.00)

Main variables and simple correlations, part 3.

Table 4

Regressions of stock price synchronicity on economy structural variables

Estimated coefficients from ordinary least squares regressions of stock price synchronicity variables, Υ_j and Ψ_j, on the logarithm of per capita GDP and structural variables. A control for market size, log(number of stocks), is included in all regressions.The structural variables are log(geographical size), variance of GDP growth, industry Herfindahl index, and the firm Herfindahl index. Regressions 4.2 and 4.4 include, as an additional structural variable, the earnings co-movement index. Sample is 37 countries, except for regressions on the earnings co-movement index, which is available for only 25 counties. Numbers in parenthesis are probability levels at which the null hypothesis of zero correlation can be rejected in two-tailed t-tests.

Dependent variable	Ψ_j is a logistic transformation of the average fraction of stocks moving together		Υ_j is a logistic transformation of the R_j^2s of regressions of stock returns on market indices	
	4.1	*4.2*	*4.3*	*4.4*
Intercept	4.36 (0.00)	8.11 (0.00)	4.66 (0.04)	8.04 (0.04)
Logarithm of per capita GDP	– 0.189 (0.01)	– 0.288 (0.01)	– 0.238 (0.04)	– 0.324 (0.04)
Logarithm of number of stocks listed	– 0.180 (0.11)	– 0.200 (0.14)	– 0.270 (0.13)	– 0.367 (0.09)
Logarithm of geographical size	– 0.867 (0.04)	– 1.89 (0.02)	– 0.948 (0.16)	– 1.78 (0.15)
Variance in GDP growth	68.6 (0.84)	– 253 (0.48)	228 (0.67)	– 106 (0.85)
Industry Herfindahl index	– 2.37 (0.27)	– 4.30 (0.09)	– 2.08 (0.54)	– 5.70 (0.15)
Firm Herfindahl index	– 0.446 (0.83)	1.49 (0.55)	– 3.71 (0.26)	1.04 (0.80)
Earning co-movement index	—	0.375 (0.49)	—	1.09 (0.22)
F statistic for the regression	3.88 (0.01)	3.51 (0.02)	2.87 (0.03)	2.50 (0.06)
Sample size	37	25	37	25
R^2	0.44	0.59	0.36	0.51

Synchronicity and economy structural variables.

Read:

- The final part of Table 3 shows correlations among structural and institutional variables.
- Table 4 asks whether structural variables explain the income-synchronicity relation. They do not: per capita GDP remains negative and significant in the synchronicity regressions.

Economic message: High synchronicity is not simply because poorer countries are smaller, more volatile, less diversified, or more concentrated.

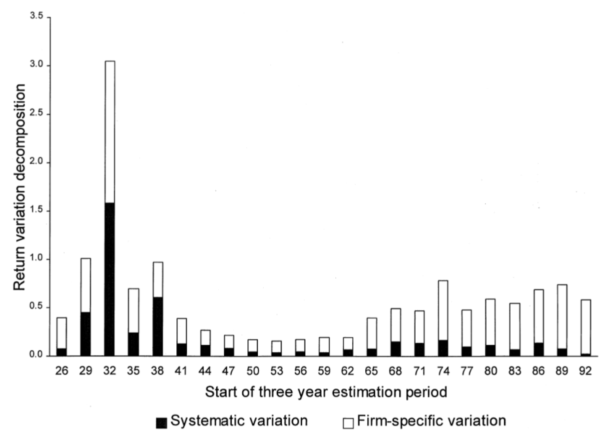
8.7 Table 5 and Figure 6: good government and U.S. variance decomposition

Table 5
Regressions of stock price synchronicity on economy structural variables and a good government index

Estimated coefficients from ordinary least squares regressions of stock price synchronicity variables, Υ_j and Ψ_j, on the logarithm of per capita GDP, structural variables and a good government index. A control for market size, log(number of stocks), is included in all regressions. The structural variables are log(geographical size), variance of GDP growth, industry Herfindahl index, and the firm Herfindahl index. Regressions 5.2 and 5.4 include, as an additional structural variable, the earnings co-movement index. Sample is 37 countries, except for regressions on the earnings co-movement index, which is available for only 25 countries. Numbers in parenthesis are probability levels at which the null hypothesis of zero correlation can be rejected in two-tailed t-tests.

Dependent variable	Ψ_j and is a logistic transformation of the average fraction of stocks moving together		Υ_j, is a logistic transformation of the R_j^2s of regressions of stock returns on market indices	
	5.1	5.2	5.3	5.4
log(GDP per capita)	0.033 (0.83)	0.025 (0.87)	0.170 (0.48)	0.188 (0.46)
Logarithm of number of stocks listed	− 0.204 (0.06)	− 0.197 (0.10)	− 0.315 (0.07)	− 0.331 (0.07)
Good government index	− 0.059 (0.11)	− 0.098 (0.03)	− 0.110 (0.07)	− 0.161 (0.03)
Strucrual variables				
Log. of geographical size	− 0.811 (0.05)	− 1.98 (0.01)	− 0.846 (0.19)	− 1.92 (0.09)
Variance in GDP growth	72.1 (0.82)	− 216 (0.49)	235 (0.65)	− 47.2 (0.93)
Industry Herfindahl index	− 3.44[a] (0.12)	− 4.97 (0.03)	− 4.07 (0.24)	− 6.79 (0.06)
Firm Herfindahl index	0.282 (0.89)	1.83 (0.41)	− 2.38 (0.46)	1.59 (0.65)
Earning comovement index	—	0.117 (0.81)	—	0.671 (0.40)
F-test for the structural variables	1.91 (0.14)	2.93 (0.05)	1.67 (0.19)	1.78 (0.17)
F statistics for the regression	3.89 (0.00)	4.63 (0.00)	3.19 (0.02)	3.55 (0.01)
Sample size	37	25	37	25
R^2	0.48	0.70	0.44	0.64

Synchronicity, structural controls, and good government.



'ig. 6. The changing structure of U.S. stock return variation from 1926 to 1995 Total returns ariation is decomposed into a systematic component, which is related to the value-weighted market ıdex, and a firm-specific component, which is not. Each bar represents an estimate over a three year eriod based on monthly dividend-adjusted returns. Returns and indexes include dividends. Data re from the Center for Research in Securities Prices.

Changing structure of U.S. stock return variation.

Read:

- Table 5 is the key institutional result: the good government index is negatively related to synchronicity, while per capita GDP becomes insignificant once good government is included.
- Figure 6 decomposes U.S. return variation into systematic and firm-specific components over time. The decline in U.S. synchronicity reflects both lower market-wide variation and more firm-specific variation.

Economic message: Institutional quality, not income by itself, is the paper's preferred explanation for why some markets have less synchronous stock prices.

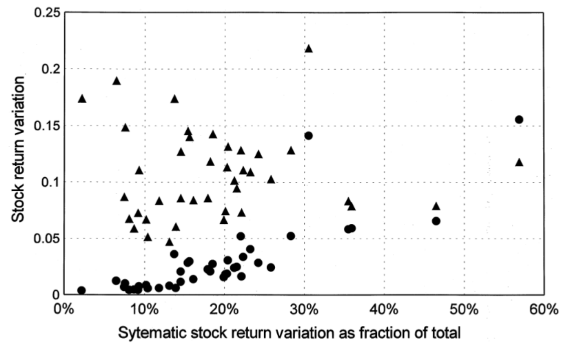
8.8 Table 6 and Figure 7: systematic versus firm-specific variation

Table 6
The components of stock return variation

Countries are ranked by stock market synchronicity, measured as the average R^2 of firm-level regressions of bi-weekly stock returns on local and U.S. market indexes in each country in 1995. This is decomposed into a firm-specific stock return variation, σ_ε^2, and a market-wide stock return variation, σ_m^2. Returns include dividends and are trimmed at $\pm 25\%$. Due to rounding errors, R_j^2 does not exactly match $\sigma_m^2/(\sigma_m^2 + \sigma_\varepsilon^2)$.

Country	R_j^2	σ_ε^2	σ_m^2	Country	R_j^2	σ_ε^2	σ_m^2
United States	0.021	0.174	0.004	Korea	0.172	0.174	0.036
Ireland	0.058	0.073	0.005	Pakistan	0.175	0.140	0.030
Canada	0.062	0.190	0.013	Italy	0.183	0.073	0.016
U.K.	0.062	0.068	0.005	Czech	0.185	0.125	0.028
Australia	0.064	0.149	0.010	India	0.189	0.132	0.031
New Zealand	0.064	0.111	0.008	Singapore	0.191	0.102	0.024
Portugal	0.068	0.084	0.006	Greece	0.192	0.103	0.024
France	0.075	0.087	0.007	Spain	0.192	0.067	0.016
Denmark	0.075	0.059	0.005	South Africa	0.197	0.074	0.018
Austria	0.093	0.061	0.006	Columbia	0.209	0.095	0.025
Holland	0.103	0.051	0.006	Chile	0.209	0.086	0.023
Germany	0.114	0.067	0.009	Japan	0.234	0.111	0.034
Norway	0.119	0.086	0.012	Thailand	0.271	0.109	0.041
Indonesia	0.140	0.127	0.021	Peru	0.288	0.128	0.052
Sweden	0.142	0.084	0.014	Mexico	0.290	0.129	0.052
Finland	0.142	0.113	0.019	Turkey	0.393	0.218	0.141
Belgium	0.146	0.047	0.008	Taiwan	0.412	0.084	0.058
Hong Kong	0.150	0.118	0.021	Malaysia	0.429	0.079	0.059
Brazil	0.161	0.143	0.027	China	0.453	0.079	0.066

Components of stock return variation.



7. Average stock return variation for each country is decomposed into a systematic componen
enoted by circles, and a firm-specific stock component, σ_ε^2, denoted by triangles, and these a
ed against stock return synchronicity, as measured by systematic variation as a percent of tot
tion. Biweekly 1995 firm-level returns are regressed on local and U.S. value-weighted indexes t
truct this data. Returns and indexes are adjusted for dividends, and are from Datastream.

Average stock return variation by component.

Read:

- Table 6 ranks countries by stock market synchronicity and decomposes return variation into firm-specific and market-wide components.
- Figure 7 shows that emerging-market synchronicity is mainly associated with high systematic variation, while developed-market asynchronicity is also associated with greater firm-specific variation.

Economic message: The problem in emerging markets is not only that firms lack idiosyncratic movement. A large part of the pattern comes from unusually high market-wide price movement.

8.9 Table 7 and Table 8 part 1: property rights and investor rights

Table 7

Regressions of systematic or firm-specific stock price variation on economy structural variables and a good government index

Ordinary least-squares regressions of the logarithm of systematic stock return variation, $\log(\sigma_m^2)$, and the logarithm of firm-specific stock return variation, $\log(\sigma_\varepsilon^2)$, on the logarithm of per capita GDP, structural variables and a good government index. A control for market size, log(number of stocks), is included in all regressions. The structural variables are log(geographical size), variance of GDP growth, industry Herfindahl index, and the firm Herfindahl index. Regressions 7.2 and 7.4 include, as an additional structural variable, the earnings co-movement index. Sample is 37 countries, except for regressions on the earnings co-movement index, which is available for only 25 countries. Numbers in parenthesis are probability levels at which the null hypothesis of zero correlation can be rejected in two-tailed t-tests.

Dependent variable	$\log(\sigma_m^2)$ is the logarithm of average systematic returns variation		$\log(\sigma_\varepsilon^2)$ is the logarithm of average firm-specific returns variation	
	7.1	7.2	7.3	7.4
log(GDP per capita)	0.228 (0.38)	0.248 (0.42)	0.057 (0.63)	0.059 (0.67)
Logarithm of number of stocks listed	– 0.176 (0.34)	– 0.313 (0.17)	0.140 (0.10)	0.0500 (0.63)
Good government index	– 0.164 (0.01)	– 0.213 (0.02)	– 0.0540 (0.06)	– 0.0520 (0.17)
Structural variables				
Log. of geographical size	– 0.418 (0.54)	– 0.128 (0.92)	0.430 (0.17)	1.79 (0.01)
Variance in GDP growth	9.59 (0.99)	– 211 (0.73)	228 (0.37)	– 167 (0.55)
Industry Herfindahl index	– 3.81 (0.30)	– 6.59 (0.12)	0.256 (0.88)	0.213 (0.91)
Firm Herfindahl index	– 1.89 (0.58)	1.26 (0.77)	0.481 (0.76)	– 0.317 (0.87)
Earning co-movement index	—	0.621 (0.51)	—	– 0.051 (0.91)
F statistics for the regression	4.29 (0.00)	3.26 (0.02)	3.18 (0.01)	3.42 (0.02)
Sample size	37	25	37	25
R^2	0.51	0.62	0.43	0.63

Systematic/firm-specific variation and good government.

Table 8

Regressions of systematic or firm-specific stock price variation on economy structural and institutional development variables across countries with good government

Ordinary least squares regressions of the logarithm of systematic stock return variation, $\log(\sigma_m^2)$, and the logarithm of firm-specific stock return variation, $\log(\sigma_\varepsilon^2)$, on the logarithm of per capita GDP, structural variables and either a good government index, an index of investor rights against corporate insiders, or both. A control for market size, log(number of stocks), is included in all regressions. The structural variables are log(geographical size), variance of GDP growth, industry Herfindahl index, and the firm Herfindahl index. Regressions 8.4 and 8.8 include, as an additional structural variable, the earnings co-movement index. The sample consists of the 22 countries whose good government index is above the average for the full sample of 37 countries. Numbers in parenthesis are probability levels at which the null hypothesis of zero correlation can be rejected in two-tailed t-tests.

Dependent variable	$\log(\sigma_\varepsilon^2)$ is the logarithm maverage firm-specific returns variation				$\log(\sigma_m^2)$ is the logarithm of average systematic returns variation			
	8.1	*8.2*	*8.3*	*8.4*	*8.5*	*8.6*	*8.7*	*8.8*
log(per capita GDP)	– 0.279 (0.45)	0.058 (0.82)	0.119 (0.73)	0.172 (0.63)	0.350 (0.64)	– 0.156 (0.83)	0.546 (0.53)	0.450 (0.63)
log of number of stocks listed	0.140 (0.25)	0.040 (0.67)	0.027 (0.80)	0.026 (0.81)	– 0.294 (0.24)	– 0.188 (0.46)	– 0.350 (0.22)	– 0.349 (0.23)
Good government index	0.036 (0.68)	–	– 0.022 (0.77)	– 0.024 (0.76)	– 0.226 (0.21)	—	– 0.254 (0.19)	– 0.251 (0.22)
Anti-director Rights index	—	0.168 (0.01)	0.173 (0.02)	0.173 (0.02)	—	0.022 (0.89)	0.085 (0.60)	0.085 (0.62)

Investor rights and stock return variation, part 1.

Read:

- Table 7 shows that the good government index is negatively related to market-wide stock return variation.
- Table 8 focuses on countries with good government and asks whether public shareholder protection is associated with more firm-specific variation.

Economic message: Two institutional channels matter: general property-rights protection is related to less market-wide noise, while public investor rights are related to more firm-specific information in developed markets.

8.10 Table 8 part 2: developed economies and firm-specific information

Structural variables								
Logarithm of geographical size	0.289	0.383	0.421	0.431	– 0.439	– 0.818	– 0.374	– 0.391
	(0.47)	(0.21)	(0.22)	(0.22)	(0.59)	(0.32)	(0.66)	(0.66)
Variance in GDP growth	– 500	– 20.3	– 3.29	– 16.4	– 446	– 399	– 201	– 177
	(0.13)	(0.95)	(0.99)	(0.96)	(0.50)	(0.63)	(0.81)	(0.84)
Industry Herfindahl index	1.37	0.918	0.541	0.803	– 4.93	– 0.969	– 5.34	– 5.80
	(0.62)	(0.62)	(0.81)	(0.74)	(0.39)	(0.85)	(0.37)	(0.36)
Firm Herfindahl index	– 1.70	– 1.67	– 1.34	– 1.60	– 2.22	– 5.97	– 2.04	– 1.56
	(0.50)	– (0.33)	(0.53)	(0.47)	(0.67)	(0.21)	(0.70)	(0.78)
Earning co-movemement index	—	—	—	– 0.319	—	—	—	0.569
				(0.56)				(0.68)
Joint significance F test for the regression	1.41	3.37	2.77	2.38	1.00	0.680	0.870	0.740
	(0.28)	(0.03)	(0.05)	(0.08)	(0.47)	(0.68)	(0.57)	(0.67)
Sample size	22	22	22	22	22	22	22	22
R^2	0.41	0.63	0.63	0.64	0.33	0.25	0.35	0.36

Investor rights and stock return variation, part 2.

Read:

- The second part of Table 8 completes the developed-country regressions.
- Anti-director rights are positively related to firm-specific variation, while the same variable has little relation to market-wide variation.

Economic message: In developed markets, stronger shareholder rights appear to help prices incorporate more firm-specific information rather than merely changing aggregate volatility.

9 Short summary

Morck, Yeung, and Yu document that emerging-market stock returns are much more synchronous than developed-market stock returns. They show that this fact is not explained away by market size, country size, industry concentration, firm concentration, GDP volatility, or earnings co-movement. The key explanatory variable is institutional quality: countries with stronger protection of private property have less synchronous stock prices. The paper then decomposes return variation and argues that poor property rights are associated with more market-wide, possibly non-fundamental price variation, while stronger investor rights in developed economies are associated with more firm-specific variation.

10 Conclusion

The paper's central lesson is that stock market prices are more informative when institutions protect property rights. Weak property rights may deter informed arbitrage, allow market-wide noise to dominate, and reduce the usefulness of stock prices as signals for capital allocation. Stronger legal and political institutions appear to create an environment in which prices reflect more firm-specific information.

For empirical asset pricing, the paper makes stock return synchronicity a cross-country measure of market informativeness. For law and finance, it provides evidence that investor protection and property rights shape not only firm value and financing, but also the informational role of stock markets. For development economics, it suggests that stock markets in weak-institution countries may allocate capital less effectively because prices are less tied to firm-specific fundamentals.